

Know When to Fold: Risk-Controlled Abstention for Protein–Ligand Co-Folding under Novelty Shift

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Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, yet a correlation is not a decision rule and it is weakest on the novel pockets and chemotypes that drug discovery cares about. We build `foldgate`, a training-free layer that turns frozen co-folding confidence into a risk-controlled accept/abstain gate with a finite-sample selective-risk guarantee, and we show the guarantee collapses under training-similarity shift: a marginally compliant gate under-controls error by about twice on novel strata, and calibrating on familiar targets while deploying on novel ones violates the guarantee in every run. Measuring reliability drift, the movement of $P(\text{correct} \mid \text{confidence})$ along a novelty axis, shows the shift is concept shift, large on structure and near zero on recency. We then prove what that implies. For a threshold on a frozen score the realized target risk is the covariate-corrected reference plus an accept-region concept gap that is invariant to any training-free reweighting. So weighted conformal silently violates its own certificate, level α is unachievable at the operating coverage once the gap exceeds the remaining slack, and only in-stratum recalibration attains the floor, given stratum labels. We validate the theorem against a known conditional and on public tabular benchmarks, and repair coverage with group-conditional calibration and a label-free worst-subpopulation (CVar/DRO) certificate under multiplicity control. On a released co-folded virtual screen the pose-confidence signal the layer is calibrated on enriches actives above docking on the post-2022 novel targets (EF@1% 9.3 vs 2.0), and abstaining on unreliable poses preserves enrichment that random abstention destroys on 75% of them. Everything runs on released predictions with no GPU.

1 Introduction

Co-folding models predict a protein–ligand complex and emit confidence scores (ipTM, ranking score, PAE). A practitioner who wants to act on a predicted pose needs a decision rule that carries an error guarantee. Conformal selective prediction supplies one: accept a pose when its score clears a threshold, abstain otherwise, and certify that the error rate among accepted poses stays below a target α with high probability [7, 13]. The catch is that this guarantee assumes exchangeability, which breaks precisely where deployment matters, on ligands and pockets dissimilar to the training set.

We make four contributions, all training-free with respect to the co-folding model and all computed from released predictions on CPU. (1) We give a finite-sample selective-risk gate over frozen co-folding confidence and show its guarantee collapses under novelty shift, then measure *reliability drift* to identify the collapse as concept shift (Sec. 3). (2) We prove an impossibility and a matching achievability result characterizing what a training-free reweighting of a frozen score can control under concept shift, validated against a known conditional and on a public non-co-folding benchmark (Sec. 4). (3) We repair coverage with group-conditional calibration and a label-free worst-subpopulation CVar/DRO certificate, with explicit multiplicity control across the five models (Sec. 5). (4) We produce the downstream decision on the pose-confidence signal the gate is calibrated on: on a released co-folded screen, abstaining on unreliable poses lifts enrichment on novel targets (Sec. 6).

2 The gate

We consider a delivered top-1 pose per system with a native confidence s and a binary label $Y = 1$ iff the symmetry-corrected ligand-RMSD is at most 2 Å. The gate accepts when $s \geq \tau$. We pick the

largest accept set (smallest τ) whose selective risk is certified below α by Learn-then-Test with fixed-sequence testing [1], testing each $H_0(\tau): P(\text{error} | s \geq \tau) \geq \alpha$ with an exact binomial tail along a pre-specified coverage grid; this gives $P(\text{selective risk} \leq \alpha) \geq 1 - \delta$, finite-sample and distribution-free. We optionally replace s with a calibration-only combined score, a shallow gradient-boosted map from cheap per-prediction features (confidence, interface ipTM, ensemble spread, cross-model agreement, ligand descriptors) to $P(\text{correct})$; the co-folding model is never retrained, and novelty is excluded from the score and enters only in calibration. We evaluate five models (AF3 [11], Boltz-1, Boltz-1x, Chai-1, Protenix [12]) on Runs N’ Poses [8], 13,535 delivered poses, at $\alpha = 0.20$, $\delta = 0.10$.

3 The exchangeability break and its mechanism

The break. We stratify targets by training-set structural and chemical similarity into S_0 (familiar) through S_4 (no analog). A single global threshold hides severe per-stratum under-coverage: for AF3 the marginal realized risk of 0.177 masks 0.375 on S_3 and 0.427 on S_4 . The operational failure is starker. When we calibrate the gate on familiar targets and deploy it on the novel region, the realized error is 0.55 while the target is 0.20, and the finite-sample guarantee holds in 0 of 100 resampled runs across all models. Marginal validity is not enough, which is the whole point.

Why it breaks. Weighted conformal would fix this if the shift were pure covariate shift, where the map $P(Y | s)$ is stable and only the marginal $P(s)$ moves. We test that assumption directly by measuring *reliability drift* $D(v) = \sum_{\text{bins}} w[P(Y | s, S_0) - P(Y | s, S_v)]$, the confidence-conditional correctness gap between the familiar reference and stratum v . Drift is large and its 90% bootstrap interval excludes zero for every model on structural novelty (Protenix reaches +0.63 on pocket novelty, Chai +0.52); on temporal novelty it is an order of magnitude smaller ($|D| \leq 0.07$, with only 2 of 15 temporal cells statistically resolved). Confidence is optimistic on novel structure, the signature of concept shift. A signed accept-region decomposition makes this quantitative: for AF3 on S_3 the target risk gap of 0.328 splits into a concept term of 0.309 (90% CI [0.27, 0.35]) and a covariate term of 0.018, which the next section turns into a theorem.

4 What a training-free reweighting can and cannot fix

The decomposition above is an instance of a general fact. Fix the frozen score s and the accept rule $\{s \geq \tau\}$. Let $\eta_P(x) = P(Y=1 | X=x)$ and η_Q be the error conditionals on the calibration law P and the deployment law Q , and call a reweighting *training-free* if its weights are covariate-measurable and use no target labels (density-ratio, classifier, or KLIEP weights, including the point mass of weighted conformal). Write $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P | s \geq \tau_c]$ for the covariate-corrected reference at coverage c and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P | s \geq \tau_c]$ for the accept-region concept gap.

Theorem (impossibility, informal; App. A). At fixed coverage c the realized target selective risk is $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$, and $\bar{\Delta}_c$ is *invariant* to the choice of training-free weights. Hence (i) the oracle-weighted plug-in certificate reports R_{ref} and still under-states the realized risk by $\bar{\Delta}_c$; and (ii) level α is unachievable at coverage c by any training-free reweighting whenever $\bar{\Delta}_c > \alpha - R_{\text{ref}}(\tau_c)$.

The proof (App. A) is a tower-property identity: a covariate-measurable weight recovers only the *source* conditional η_P and cannot reach η_Q , which completes the covariate-shift boundary of weighted conformal [3] and specializes the irreducible term of domain-adaptation lower bounds [15]. Matching achievability: in-stratum recalibration attains the floor R_{Q_g} given n_g stratum labels, with slack $O(\sqrt{\log(1/\delta)/n_g})$; pointwise conditional coverage being impossible [6], the guarantee is neighborhood-conditional.

Validation on a general benchmark. A synthetic generator with a known $P(Y | s, v)$ gives ground truth for each claim. With genuine target drift, $R_{\text{ref}} = 0.273$ and $\bar{\Delta} = 0.205$ sum to the realized $R_Q = 0.478$, the realized risk is invariant across nine reweightings (residual < 0.001), the oracle certificate under-reports by $\bar{\Delta}$, and above the crossing no reweighting reaches α ; in-stratum recal-

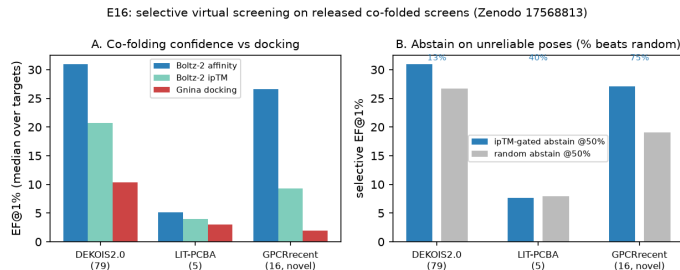


Figure 1: Selective virtual screening on released co-folded screens. (A) The pose signal (ipTM) the layer governs enriches actives above docking, with the ungoverned Boltz-2 affinity head shown for comparison; the gap is largest on the post-2022 novel GPCR targets. (B) Ranking by the affinity head and abstaining on low-ipTM poses at 50% coverage preserves enrichment that random abstention destroys; percentages are the fraction of targets whose selective EF beats the random-abstention 95th percentile.

Table 1: Selected results. AURC is out-of-fold under target-grouped (leakage-free) cross-validation; m^* is the smallest accepted-mass fraction certified at α by the label-free CVaR bound. † Chai native ranking certifies almost no coverage alone, an honest limit the combined score recovers.

	AF3	Boltz-1	Boltz-1x	Chai-1	Protenix
AURC native $\rightarrow$ combined	.185 $\rightarrow$.118	.234 $\rightarrow$.162	.215 $\rightarrow$.154	.247 $\rightarrow$.146	.281 $\rightarrow$.166
LOTO realized risk ($\leq .20$)	.185	.157	.168	.30 †	.191
reliability drift D , pocket S_3	+.47	+.47	+.47	+.52	+.63
coverage to certify $m^* \leq .5$	.45	.20	.25	.40	.25

ibration controls only with target-stratum labels, and an error floor above α forces abstention. A shared-label variant reproduces the decomposition and invariance with a near-zero pooled gap, placing the concept effect at the worst stratum as in the co-folding break. On public benchmarks with no co-folding, a single-threshold gate under-controls the worst novelty block, weighted conformal issues a certificate its realized risk then violates, and group-conditional calibration certifies control by abstaining on the blocks it cannot cover: on elec2 (worst-block concept gap 0.36) the realized worst-block risks are 0.31, 0.34, and 0.20, and on ACS Income (covariate gap 0.05) weighted conformal already repairs the worst state (0.23 $\rightarrow$ 0.20).

5 Repairing the guarantee

Group-conditional calibration keys a separate threshold per novelty stratum, restoring per-stratum control at an honest coverage cost (with the combined score AF3 accepts 52% of S_1 and 39% of S_2 at risk $\leq \alpha$, against 8% and 12% natively). It needs stratum labels, which Runs N’ Poses ships.

A label-free worst-subpopulation certificate. When the novel subpopulation is unnamed, we control the conditional value-at-risk of the accepted loss. For the binary label, $\text{CVaR}_\beta = \min(1, r/(1 - \beta))$ with accepted error rate r , so a finite-sample upper bound on r [2] certifies $\text{CVaR}_\beta \leq \alpha$ for every β at once [4]. By CVaR–DRO duality this certifies that every accepted subpopulation of mass at least $m^* = 1 - \beta^*$ has error $\leq \alpha$, with no stratum labels and no importance weights. The combined score certifies $m^* \leq 0.5$ (every half-subpopulation) at 20–45% coverage where the native score often certifies nothing, and is largely vacuous on deploy-to-novel ($m^* \rightarrow 1$ for most models), which we report. The same certificate reparameterizes to a chi-square robustness radius ρ^* [14], and a Bonferroni δ/K correction yields one certificate jointly across all five models (at 20% coverage, $m^* = 0.52$ and $\rho^* = 0.10$ for every model at once).

Multiplicity across models. Joint certificate claims (the gate, leave-one-target-out validity, the CVaR/DRO bounds) take a Bonferroni union bound at $\delta/K = 0.02$; the joint leave-one-target-out gate then holds for AF3, Boltz-1, and Boltz-1x with real coverage while Chai and Protenix satisfy it

only by abstaining. "For every model" significance conjunctions are intersection-union tests, valid without penalty. The 55-cell drift grid uses a Romano-Wolf step-down bootstrap [17] at family-wise 0.10; 42 cells survive, all structural ones among them, while the temporal cells fail a 0.05-margin equivalence test, leaving only the temporal-versus-structural magnitude contrast supported.

Weighted conformal serves as a label-free complement that abstains on real novel data (Sec. 4), so group-conditional calibration stays the operative guarantee. A training-free surrogate that keys the gate on the model’s own ensemble disagreement recovers control on moderate novelty but stays a weak proxy on the extreme.

6 Utility and a downstream decision

Ranking utility, validated without leakage. The combined score dominates native confidence on the risk-coverage curve, cutting its area by 28–41% (AF3 0.185 $\rightarrow$ 0.118, Protenix 0.281 $\rightarrow$ 0.166), about two to three times more poses retained at a fixed guarantee. Because a target recurs across models, a naive pooled split leaks 95.1% of test targets into calibration; under target-grouped cross-validation the gate still holds (AF3 risk 0.185, [0.168, 0.203], 5/5 folds) and the advantage survives (Table 1).

The decision number. We reuse a released co-folded virtual screen [10] with per-compound confidence and active/decoy labels for DEKOIS2.0 (79 targets), LIT-PCBA (5), and GPCRrecent (16 post-2022 novel targets), with no GPU and no crystal coordinates. We report the pose confidence (ipTM) the reliability layer governs and keep the Boltz-2 affinity head as an ungoverned comparator (Fig. 1). The governed pose signal enriches actives above docking, and the gap widens on novelty: EF@1% is 20.7 against Gnina 10.3 on DEKOIS, and 9.3 ([5.7, 14.8]) against Gnina 2.0 and Glide 3.8 on the post-2022 GPCR targets, an interval disjoint from docking’s [1.5, 2.6]; the affinity head reaches 26.6 but carries no guarantee, and AF3’s pose ipTM (25.8 on DEKOIS) only widens the pose-over-docking gap. For selective screening we rank by the affinity head and abstain on low-ipTM poses, so the layer supplies the abstention while the affinity head supplies the ranking, preserving enrichment that random abstention destroys on 75% of the novel GPCR targets against 13% on saturated DEKOIS. The layer, calibrated on pose correctness, transfers here as a heuristic abstention rule without a coverage guarantee, since the screen carries no pose-RMSD label. Enrichment degrades on novel actives: the affinity head trends from 27 on the most novel similarity bin to 42 on train-similar ($n \approx 16$, CIs overlap), and the governed pose signal sits lower on novel actives (11) than familiar (18), the screening echo of Sec. 3. A Murcko-scaffold split of the DEKOIS actives shows enrichment lower on novel scaffolds for every method, recomputed rather than inherited; because the released decoys ship no structures we cannot correct for analog bias, so absolute EF may be inflated and the pose-over-docking gap and novelty trend carry the claim.

7 Cross-dataset generality and limitations

A threshold frozen on Runs N’ Poses and deployed on FoldBench [9] holds for AF3 ($0.15 \leq \alpha$) and under-controls for Protenix (0.25), an honest transfer limited by the confidence fields FoldBench releases. The break is structural, and the theorem and its repair generalize beyond co-folding to the public elec2 and ACS benchmarks (Sec. 4). The no-analog stratum ($n \approx 72$ per model) is un-certifiable under every method, where the only action is abstention, and a randomly-localized gate keyed continuously on novelty [5] still leaves that tail. The co-folding guarantee itself rests on a single labeled dataset (Runs N’ Poses), since FoldBench contributes one shared field and the public benchmarks are tabular analogues; a feature-matched cross-dataset positive and a prospective screen validation are the main reach items.

8 Conclusion

Co-folding confidence becomes a risk-controlled decision only when the guarantee is audited where it fails. That failure is concept shift on structural novelty; the paper proves which repairs it admits and applies group-conditional and label-free methods that need no training-set access, all on released

predictions with no GPU. The code is public.

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A Proof of the impossibility, and the achievability bracket

Setup. Frozen score $s = s(X)$, novelty $v = v(X)$, error label $Y \in \{0, 1\}$, accept region $A_\tau = \{s \geq \tau\} \in \sigma(X)$. A training-free reweighting has weights $w = w(X) \geq 0$ that are $\sigma(X)$ -measurable and use no target labels. Assume s has no atom at the operating threshold, so for each coverage c there is a unique τ_c with $Q(s \geq \tau_c) = c$; the real ipTM and ranking scores carry ties, handled by boundary randomization, under which the equalities below become $\leq / \geq$ brackets.

Lemma (reweighting sees only the source labels). For any such w , the reweighted plug-in selective risk is $\tilde{E}^w(\tau) = \mathbb{E}_P[wY \mathbf{1}_{A_\tau}] / \mathbb{E}_P[w \mathbf{1}_{A_\tau}] = \mathbb{E}_{Q_w}[\eta_P | A_\tau]$, with $dQ_w \propto w dP_X$. *Proof.* w and $\mathbf{1}_{A_\tau}$ are $\sigma(X)$ -measurable, so tower over X : $\mathbb{E}_P[wY \mathbf{1}_{A_\tau}] = \mathbb{E}_P[w \eta_P \mathbf{1}_{A_\tau}]$; normalize. The target conditional η_Q cannot appear. $\square$

Theorem. At coverage c : (a) $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$ with $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P | A_{\tau_c}]$ and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P | A_{\tau_c}]$; (b) any w realizing coverage c accepts exactly A_{τ_c} , so its realized risk equals $R_Q(\tau_c)$, independent of w ; (c) with

oracle weights the certificate of the Lemma equals $R_{\text{ref}}(\tau_c)$, so it under-reports the realized risk by exactly $\bar{\Delta}_c$; (d) level α is achievable at coverage c by some training-free reweighting iff $R_Q(\tau_c) \leq \alpha$, i.e. $\bar{\Delta}_c \leq \alpha - R_{\text{ref}}(\tau_c)$. *Proof.* (a) $A_{\tau_c} \in \sigma(X)$, so $R_Q(\tau_c) = \mathbb{E}_Q[\eta_Q | A_{\tau_c}]$; add and subtract $\mathbb{E}_Q[\eta_P | A_{\tau_c}]$. (b) The no-atom assumption pins τ_c from c ; realized risk is a functional of (Q, τ_c) alone. (c) Lemma with $w^* = dQ_X/dP_X$ gives $Q_{w^*} = Q$, so the certificate is R_{ref} ; subtract (a). (d) is (a)+(b). $\square$

This is coverage-pinned: it leaves open reaching α by lowering coverage, and it binds only the class of scalar thresholds of a fixed score with covariate-measurable weights. Re-scoring on v or a per-stratum threshold escapes it, which is the achievability side: in-stratum split calibration [13] gives stratum-conditional miscoverage $\leq 1/(n_g+1)$ regardless of drift, conformal risk control [16] gives the same slack for the joint accept-and-err rate, and the conditional selective risk (a ratio with a random denominator, to which the clean $1/(n_g+1)$ does not apply) is controlled with high probability by Learn-then-Test with slack $O(\sqrt{\log(1/\delta)/n_g})$. Per stratum the impossibility floor and the achievability ceiling meet at $R_{Q_g}(\tau_c)$, the frozen score's own in-stratum accept-region risk, reachable by stratified recalibration and provably not by covariate reweighting. The label-free ceiling of the DRO/CVaR certificate matches the worst case over the assumed ambiguity ball and is the honest fallback when no stratum labels are available, vacuous exactly when the ball fails to cover the true drift, as it does on deploy-to-novel.

B Benchmark and validation details

The synthetic generator draws a stratum k from a tilted v -marginal, a score $z \sim \mathcal{N}(m_0 - Ev_k, \sigma^2)$, and a label $y \sim \text{Bernoulli}(\pi)$ with $\pi = \varepsilon + (1 - 2\varepsilon) \text{sigmoid}(\beta_0 + \beta_1 z - D_{\text{eff}} v)$, where $D_{\text{eff}} = D$ on the calibration law and $D + D_q$ on the target law. The knobs isolate covariate shift (E , tilt), genuine concept drift (D_q), and an irreducible error floor (ε); every theorem quantity is available in closed form by quadrature and cross-checked against a Monte-Carlo draw. The real benchmarks reduce any classifier to the triple (s, y, v) with s its confidence, $y = \mathbf{1}[\text{correct}]$, and v the natural shift axis (temporal block for elec2, target state for ACS Income); the base classifier is a gradient-boosted tree, torch-free, and the split is grouped by the exchangeable stratum. Realized worst-stratum selective risk at $\alpha = 0.20$: elec2 marginal 0.31, weighted-conformal 0.34, group-conditional 0.20; ACS Income marginal 0.23, weighted-conformal 0.20, group-conditional 0.20. The reliability drift that separates the two cases is 0.36 (elec2, concept) against 0.05 (ACS, covariate).